

SECURITY INCIDENT REPORT

CyberLab Azure Security Operations Center

INC-2026-001 | June 17-18, 2026

Incident ID	INC-2026-001
Date/Time Detected	June 17, 2026 @ 22:35 UTC
Date/Time Resolved	June 18, 2026 @ 02:50 UTC
Severity	MEDIUM
Status	RESOLVED
Analyst	Axcel Habon — CyberLab SOC
Affected Assets	web-1 (10.0.1.4), web-2 (10.0.1.5)
Attacker IP	10.0.0.4 (JumpBox — internal pivot simulation)

1. Executive Summary

On June 17, 2026, the CyberLab Security Operations Center detected a coordinated reconnaissance and vulnerability scanning attack targeting two web servers (web-1 and web-2) hosted on Microsoft Azure. The attack originated from internal IP 10.0.0.4 (JumpBox — simulating an external attacker pivot). Both servers were simultaneously targeted using automated scanning tools.

Investigation confirmed that while the attacker successfully probed exposed web pages, no sensitive files were accessed, no credentials were compromised, and no unauthorized access to the system was achieved. The attacker IP was blocked via Azure NSG rules following the investigation.

2. Detection & Timeline

Time (UTC)	Event
Jun 17 @ 22:35	Spike detected on web-1 — 1,100 events in under 5 seconds
Jun 17 @ 22:46	Nikto 2.1.5 scanner identified in logs — automated vulnerability scan initiated
Jun 17 @ 22:46	web-2 simultaneously targeted — coordinated scan confirmed
Jun 18 @ 02:46	Directory traversal attempts detected — /etc/passwd, /etc/shadow targeted
Jun 18 @ 02:46	Hydra HTTP brute force against DVWA login page detected
Jun 18 @ 02:50	Investigation complete — attacker IP blocked in Azure NSG

3. Attack Analysis

3.1 Attack Tools Identified

- Nikto v2.1.5 — Web vulnerability scanner (User-Agent confirmed in logs)
- Hydra v9.5 — HTTP POST brute force tool against DVWA login page
- Nmap — Port scanning tool used for initial reconnaissance

3.2 Attack Techniques (MITRE ATT&CK)

Technique ID	Technique Name	Evidence
T1595.001	Active Scanning	Nmap scan — ports 22, 80 identified
T1190	Exploit Public-Facing App	Nikto probing DVWA for known vulnerabilities
T1110.001	Brute Force — Password Guessing	Hydra HTTP POST to /login.php with 10 password attempts
T1083	File & Directory Discovery	Directory traversal attempts to /etc/passwd, /etc/shadow

4. Investigation Findings

4.1 Key Evidence

Log evidence from ELK Stack (Kibana) container logs:

```
[Thu Jun 18 02:46:12] [client 10.0.0.4:38290] AH00126: Invalid URI in request GET /help/../../../../../../etc/shadow HTTP/1.1
```

Apache correctly rejected the directory traversal attempt with error AH00126 — Invalid URI.

4.2 Vulnerability Findings (Nikto)

- Cookie PHPSESSID created without httponly flag — session hijacking risk
- Missing X-Frame-Options header — clickjacking vulnerability
- /config/ directory indexing exposed — configuration data visible
- Apache version disclosed in server headers — version fingerprinting possible
- Apache/2.4.25 (Debian) — outdated version with known CVEs

4.3 Breach Assessment

Check	Result	Evidence
Was /etc/passwd accessed?	NO	404 response

Was /etc/shadow accessed?	NO	AH00126 error
Were credentials compromised?	NO	SSH key-only auth
Was DVWA admin account breached?	NO	DVWA intentionally vulnerable
Was lateral movement detected?	NO	No internal traffic post-scan

5. Response Actions Taken

5.1 Containment

- Blocked attacker IP 10.0.0.4 in web-1-nsg (Inbound Deny rule, Priority 100)
- Blocked attacker IP 10.0.0.4 in web-2-nsg (Inbound Deny rule, Priority 100)

5.2 Evidence Collection

- ELK Stack container logs preserved — 14,070+ events captured
- Kibana SOC Dashboard created showing attack timeline
- Microsoft Sentinel connected and monitoring Azure activity
- Filebeat Apache and container modules enabled for future detection

6. Recommendations

- Update Apache from version 2.4.25 to latest stable — multiple CVEs present in current version
- Enable httponly and Secure flags on all cookies
- Add X-Frame-Options and Content-Security-Policy headers
- Disable /config/ directory indexing in Apache configuration
- Implement rate limiting on login endpoints to prevent brute force
- Enable Microsoft Defender for Cloud on all VMs
- Configure Sentinel analytics rules for automated brute force detection
- Deploy Azure Web Application Firewall (WAF) in front of web servers

7. Lessons Learned

This incident demonstrated the importance of multi-layered security monitoring. The ELK Stack detected the attack at the application layer while Microsoft Sentinel monitored the Azure platform layer — both SIEMs provided complementary visibility.

Key takeaway: A single SIEM is not sufficient in enterprise environments. The combination of ELK (application logs) and Sentinel (cloud platform logs) provided complete attack visibility from initial reconnaissance through containment.

Analyst	Date Closed
Axcel Habon — SOC Analyst	June 20, 2026